A Project Synopsis Submitted for

Bachelor of Technology

Project & Seminar (7SA08)



PROJECT TITLE	**: CareBot AI — Your Intelligent Healthcare Companion**
PROJECT GROUP NO.	**: 17**
NAME OF PROJECTEE'S	**: 1) Vidhi Vijay Ingle** **2) Sayali Dinesh Deshmukh** **3) Sujal Shriram Jadhav** **4) Pratik Vilas Wankhede**
NAME OF GUIDE	**: Prof. R.C. Awtade**

Department of Artificial Intelligence & Data Science,

Prof. Ram Meghe Institute of Technology & Research, Badnera- Amravati.

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Project Title: CareBot AI — Your Intelligent Healthcare Companion

Abstract:

Timely access to reliable, understandable health information is a persistent challenge, particularly at the stage before a patient consults a doctor. CareBot AI is a Generative AI-powered healthcare copilot that integrates Machine Learning, Deep Learning, Computer Vision, Natural Language Processing and Generative AI into a single, unified web-based platform to address this gap. The system allows a user to enter symptoms and receive AI-based disease predictions with confidence scores, upload chest or bone X-rays for Convolutional-Neural-Network-based analysis of conditions such as pneumonia and fractures, and upload medical reports such as blood tests, CBC or MRI reports for automated extraction and explanation in simple language using Optical Character Recognition together with Generative AI. The platform also provides an AI-driven conversational health assistant, a personalized diet plan generated from age, weight, BMI and diagnosed condition, specialist-doctor recommendations, medicine reminders, and a consolidated health dashboard that tracks a user's health history over time. Unlike general-purpose conversational AI tools, CareBot AI combines specialized healthcare models with Generative AI explanations, and unlike existing single-purpose health applications, it unifies symptom analysis, medical imaging, report interpretation and personalized guidance into one ecosystem. The system is implemented using Python and Flask for the backend, MySQL for storage, TensorFlow/Keras and OpenCV for deep learning and computer vision, EasyOCR/Tesseract for text extraction, and the Gemini API for generative explanations, with an HTML, CSS, JavaScript and Bootstrap frontend. The proposed system is intended strictly as a decision-support and health-awareness tool rather than a diagnostic replacement, and it consistently directs users to consult qualified medical professionals. This project demonstrates the practical integration of multiple AI domains into a socially relevant healthcare application.

I. Introduction:

Artificial Intelligence has rapidly become a driving force in modern healthcare, enabling faster and more accessible preliminary health guidance than was previously possible. However, most existing digital health tools address only a single function — either a rule-based symptom checker, a general-purpose chatbot, or a standalone medical-image classifier — and rarely combine these capabilities into one coherent platform. CareBot AI is proposed as a Generative-AI-powered healthcare copilot that unifies symptom analysis, chest and bone X-ray interpretation, medical-report reading, an AI health chat assistant, personalized diet guidance, doctor recommendations, medicine reminders and a health analytics dashboard within a single web-based system. The platform is built on a layered architecture in which a Machine Learning layer performs disease prediction from reported symptoms, a Computer Vision layer (Convolutional Neural Networks) analyses uploaded X-ray images, an OCR layer extracts text from uploaded medical reports, and a Generative AI layer (the Gemini API) converts the technical outputs of these models into clear, patient-friendly explanations and conversational responses. By combining specialized, trained healthcare models with the explanatory power of Generative AI, CareBot AI aims to go beyond both a basic symptom checker and a general-purpose assistant such as ChatGPT, neither of which maintains patient history, analyses medical images directly, or produces personalized, model-driven health recommendations.

a. **Motivation (Problem Statement):** Patients frequently face difficulty in understanding their symptoms, interpreting medical reports full of technical terminology, and knowing which specialist to consult, especially in areas with limited immediate access to a doctor. Existing applications typically solve only one of these problems in isolation, forcing users to rely on multiple disconnected tools or on general-purpose AI assistants that are not trained on healthcare-specific data and cannot process medical images or reports directly. This fragmentation, combined with a lack of plain-language explanation for AI-generated predictions, motivates the development of a single, explainable, AI-driven healthcare copilot that can guide a user from symptom entry through report interpretation to personalized, actionable advice, while clearly stating that it supports — rather than replaces — professional medical diagnosis.

II. Feasibility Study:

Technical feasibility is high, as the required technologies — Python, Flask, MySQL, TensorFlow/Keras, OpenCV, EasyOCR/Tesseract and the Gemini API — are mature, well-documented and freely available for academic use, and public datasets (Kaggle disease-symptom datasets, the Chest X-ray Pneumonia dataset, NIH ChestX-ray14 and RSNA Pneumonia Detection data) provide sufficient training data for the prediction and image-analysis modules. Operational feasibility is favourable because the platform is delivered through a standard web browser, requiring no specialized hardware or training for end users, and its modular design allows features to be developed and tested independently before integration. Economic feasibility is strong since the project relies on open-source frameworks and free-tier or low-cost generative-AI API access, keeping development cost within the scope of a final-year academic project. The main risks — variable X-ray image quality, the general-purpose (non-diagnostic) nature of the AI predictions, and dependence on third-party generative-AI API availability — are manageable within the project timeline and are mitigated by displaying confidence scores, medical disclaimers, and encouraging professional consultation for all AI-generated results.

III. Literature Survey/Related Work:

Artificial Intelligence has significantly transformed healthcare by enabling intelligent disease prediction, medical-image analysis, and clinical decision support. Recent advances in Generative AI, Machine Learning, Natural Language Processing and Computer Vision have further extended these capabilities. The following studies form the foundation for the proposed CareBot AI system.

[1] Brown et al. (2020) – Language Models are Few-Shot Learners. Introduced GPT-3, showing large language models can perform conversational tasks with minimal training. The model is general-purpose and lacks healthcare-specific functionality, but forms the conceptual basis for CareBot AI's Generative-AI layer.

[2] Rajpurkar et al. (2017) – CheXNet: Radiologist-Level Pneumonia Detection. Proposed CheXNet, a CNN detecting pneumonia from chest X-rays at radiologist-level accuracy. It performs detection only, without personalized recommendations or plain-language explanation — a gap CareBot AI fills via its Generative-AI layer.

[3] Singhal et al. (2023) – Large Language Models Encode Clinical Knowledge. Developed Med-PaLM, a healthcare-focused LLM answering medical questions with clinically relevant responses. It does not integrate medical-image analysis, report interpretation and patient history into one platform, as CareBot AI proposes.

[4] Devlin et al. (2019) – BERT: Pre-training of Deep Bidirectional Transformers. Introduced BERT, improving contextual text understanding via bidirectional pre-training, widely used for symptom interpretation and document analysis, though additional models are still needed for image-based prediction.

[5] He et al. (2016) – Deep Residual Learning for Image Recognition. Proposed the ResNet architecture using residual connections to train very deep networks; ResNet-based CNNs are widely used for medical-image classification and are considered for CareBot AI's X-ray analysis module.

[6] World Health Organization (2021) – Ethics and Governance of AI for Health. Recommends that healthcare AI be transparent, fair, privacy-preserving and accountable, and should assist rather than replace healthcare professionals — principles guiding CareBot AI's design as a decision-support system.

Research Gap: Most existing healthcare applications address only one function — symptom checking, image classification, or conversational AI — in isolation. Very few systems combine Generative AI, Machine Learning, Computer Vision and NLP for symptom analysis, X-ray analysis, medical-report interpretation and personalized guidance within a single integrated platform, and many existing systems provide predictions without an accessible explanation for the end user.

Proposed Solution: CareBot AI addresses this gap by integrating Generative AI with Machine Learning, Computer Vision and NLP into one unified platform that performs symptom analysis, X-ray analysis, medical-report interpretation, AI-powered health conversation, personalized diet and doctor recommendations, medicine reminders and health-record management, while consistently presenting its output as preliminary, AI-assisted guidance rather than a medical diagnosis.

IV. Objective & Scope of the Project:

Objectives:

i) To design and develop a unified, web-based AI healthcare platform that integrates Machine Learning, Deep Learning, Computer Vision, NLP and Generative AI to provide intelligent healthcare services.

ii) To build a symptom-checker module that predicts probable conditions from user-reported symptoms with an associated confidence score.

iii) To develop an intelligent medical image and report analysis module that utilizes **CNN, OCR, and Generative AI** to analyze uploaded **X-ray images** and **medical reports**, and present the results in simple, user-friendly language.

iv) To provide personalized healthcare support through an AI chatbot, including health guidance, diet recommendations, doctor-specialization suggestions, and medicine reminders.

v) To develop a secure health dashboard that stores, manages, and visualizes users' medical history, BMI, and health trends for continuous health monitoring.

vi) To ensure the system provides reliable decision-support recommendations while clearly stating that it is intended to assist users and not replace professional medical diagnosis or treatment.

Scope:

The scope of the project covers the design and development of a web-based platform offering symptom-based disease prediction, X-ray image analysis, medical-report interpretation, an AI health-chat assistant, personalized diet recommendations, doctor recommendations, medicine reminders and a health analytics dashboard, using publicly available datasets for model training. Real-time hospital integration, clinical deployment, and wearable/smartwatch data integration (heart-rate, SpO2 and sleep monitoring) are identified as future scope beyond the current academic implementation.

V. Requirement for the proposed work:

Software Requirements: Python 3.x, Flask, MySQL, TensorFlow/Keras, OpenCV, Scikit-learn, EasyOCR/Tesseract, Gemini API, HTML5, CSS3, JavaScript, Bootstrap, Chart.js, Git/GitHub, Visual Studio Code, Windows/Linux operating system.

Hardware Requirements: A system with a minimum of Intel i5 processor (or equivalent), 8 GB RAM (16 GB recommended for model training), 256 GB storage, and a dedicated GPU (e.g., NVIDIA GTX/RTX series) recommended for training deep-learning models; standard internet connectivity for API access.

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Students Name	**Signature**
1. Vidhi Vijay Ingle	_____
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Signature

(Prof. R.C. Awtade)